

SEMEN QUALITY IN RELATION TO ANTIOXIDANT INTAKE IN A HEALTHY MALE POPULATION

Objective To assess the relationship between dietary antioxidant intake and semen quality in young healthy males **Design** Cross-sectional study **Setting** University and college campuses in the Rochester, New York, area **Patients** 189 university-aged men **Interventions** None **Main Outcome Measures** Semen volume, total sperm count, concentration, motility, total motile count, and morphology **Results** Progressive motility was 6.5 (95% CI 0.6, 12.3) percentage units higher among men in the highest quartile of β -carotene intake compared to men in the lowest quartile. Similar results were observed for lutein intake. Lycopene intake was positively related to sperm morphology. The adjusted percentages (95% CI) of morphologically normal sperm in increasing quartiles of lycopene intake were 8.0 (6.7, 9.3), 7.7 (6.4, 9.0), 9.2 (7.9, 10.5) and 9.7 (8.4, 11.0). There was a non-linear relationship between vitamin C intake and sperm concentration, with men in the second quartile of intake having, on average, the highest sperm concentrations and men in the top quartile of intake having the lowest concentrations. **Conclusions** In a population of healthy young men, carotenoid intake was associated with higher sperm motility and, in the case of lycopene, better sperm morphology. Our data suggest that dietary carotenoids may have a positive impact on semen quality.